

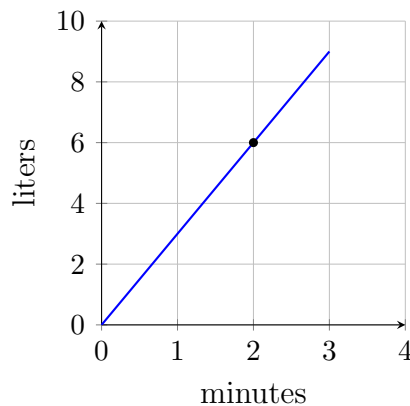
Graphing Proportional Relationships

Reading unit rates from graphs, writing equations, and graphing from tables

Directions:

- A proportional relationship graphs as a **straight line through the origin** $(0, 0)$, and its unit rate is the y -value at $x = 1$.
- Show your work beside each graph. Units are stated in each question.

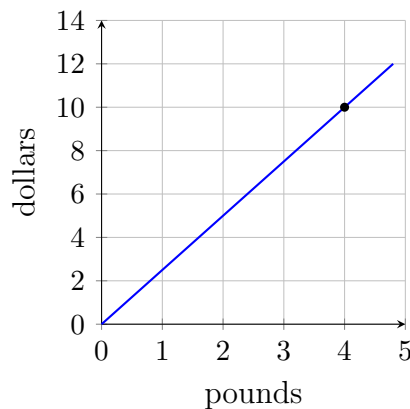
1. A hose fills a tank. The graph shows y , the liters of water in the tank, after x minutes. Find the unit rate, in liters per minute.



Work space:

Answer: _____

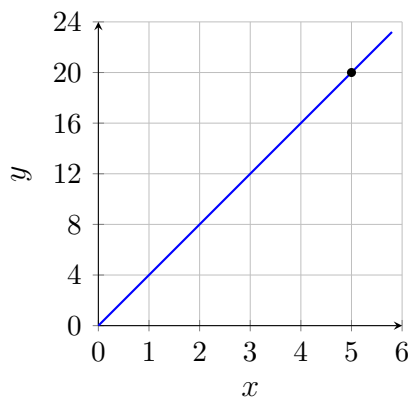
2. The graph shows the cost y , in dollars, of x pounds of grapes. Find the price per pound, in dollars.



Work space:

Answer: _____

3. The graph shows a proportional relationship. Write its equation in the form $y = kx$.

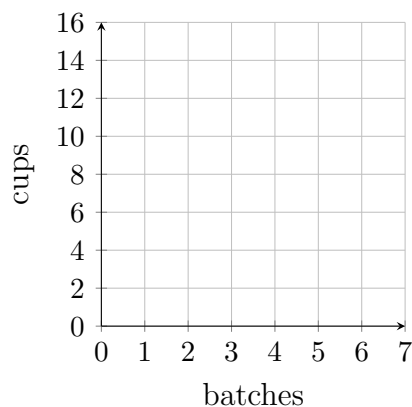


Work space:

Answer: _____

4. A recipe uses 5 cups of flour for every 2 batches of muffins. The table shows the pattern.

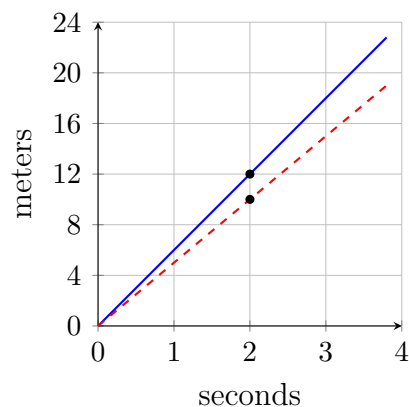
batches (x)	0	2	4	6
cups of flour (y)	0	5	10	15



(a) Plot the four points on the grid and join them with a line.

(b) Unit rate, in cups per batch:

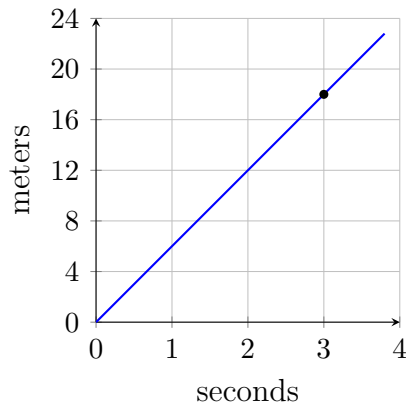
5. Two runners' distances are graphed below: y meters after x seconds. Runner A (solid) passes through $(2, 12)$ and Runner B (dashed) passes through $(2, 10)$. How much greater is Runner A's unit rate, in meters per second?



Work space:

Answer: _____

6. A remote-control car drives at a steady speed. The graph shows its distance y , in meters, after x seconds.

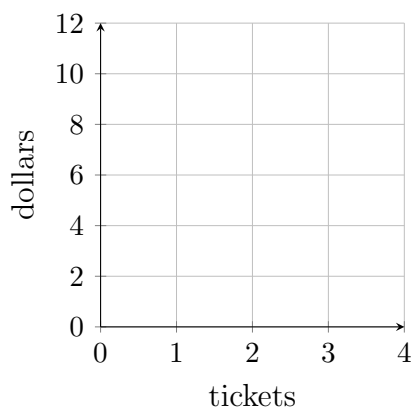


(a) Equation of the line: _____

(b) Distance, in meters, after 7 seconds: _____

7. Nia recorded the cost of raffle tickets in this table.

tickets (x)	1	2	3
cost in dollars (y)	4	8	11

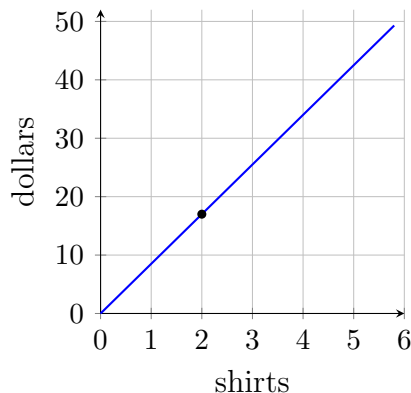


(a) Plot the three points on the grid.

(b) Cost per ticket at $x = 2$: _____
and at $x = 3$: _____

(c) Say whether the relationship is proportional and how the graph shows it.

8. A shop prints custom shirts for a constant price each. The graph shows the cost y , in dollars, of x shirts.

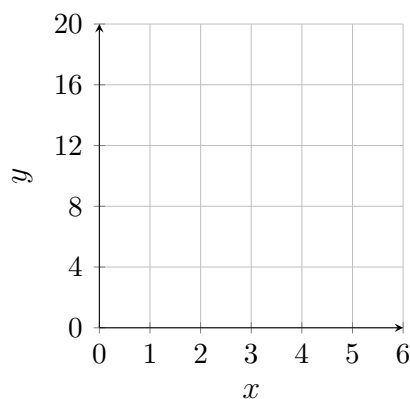


(a) Equation of the line: _____

(b) How many shirts can you buy for 51 dollars? _____

9. The table shows a proportional relationship, but two entries are missing.

x	1	_____	5
y	3.5	7	_____

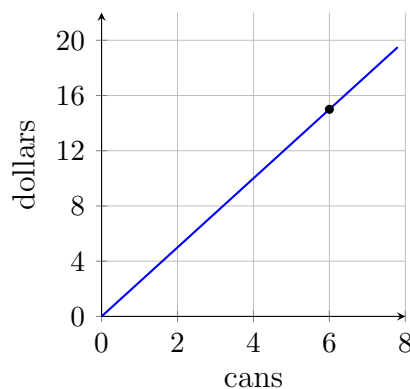


(a) Missing x -value: _____

Missing y -value: _____

(b) Plot the three completed points on the grid and join them with a line.

10. Store A's price for canned peaches is graphed below: 15 dollars for 6 cans, through the origin. Store B charges $y = 2.75x$ dollars for x cans.



(a) Compare the unit rates with $<$, $>$, or $=$:

Store A's rate _____ Store B's rate

(b) Cost, in dollars, of 12 cans at the cheaper store: _____